

42172 Introduction to Artificial Intelligence

Assessment Task 1 - Example I

Search Optimisation

Sydney inspection route using simulated annealing and a genetic algorithm

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Google Colab notebook link

https://drive.google.com/file/d/1ey94wUQnYqz3DWoyZ1HTj0tiWp54LBgd/view?usp=drive_link

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Executive Summary

This work analyzes a concise AI application that generates a route connecting twenty public places in downtown Sydney. The route should start and end in Sydney Town Hall and visit all other places only once, forming a traveling salesperson problem (TSP). Too many permutations exist to test directly, so two common algorithms were analyzed: simulated annealing (SA) and a genetic algorithm (GA).

Coordinates of each location were taken from Google Maps and confirmed before applying the algorithms. Haversine distance measure was used to calculate the distance between each pair of locations. Both algorithms used the same dataset, a single depot, and the same route-validity rules. A basic solution was developed using the nearest-neighbor heuristic, resulting in a total distance of 12.344 km. After a short parameter-tuning session, both approaches were run 30 times using random seeds.

Both algorithms discovered the same best route distance of 8.896 km. The genetic algorithm achieved an average route distance of 8.911 km (SD = 0.003), with all routes no farther than 1% from the pooled best. The simulated annealing approach provided an average value of 8.972 km (SD = 0.206), with 80% of runs being no farther than 1% from the best one. In terms of computational efficiency, the simulated annealing approach required an average of 0.085 seconds per run, while the GA needed 1.121 seconds. Based on the current results, the GA is preferable because of its much better consistency and 27.8% improvement in mean route length.

In the new data assessment, UTS Tower was added to the model as the 21st location, and both models were repeated ten times without retuning. For the simulated annealing method, the mean distance was 10.245 km (SD 0.087), and for the genetic algorithm, the mean distance was 10.295 km (SD 0.049). This means that, in this case, simulated annealing produced a lower distance and shorter computation time than the genetic algorithm, which, however, showed more consistent results.

Decision metric	SA	GA
Mean distance	8.972 km	8.911 km
Standard deviation	0.206 km	0.003 km
Runs within 1% of pooled best	80%	100%
Mean runtime	0.085 s	1.121 s

1. Introduction

Context and significance

A worker visiting different places will waste considerable time if the order of visitation is not determined as efficiently as possible. In the current example, twenty different public places located in central Sydney form one route which starts and ends at Sydney Town Hall. At first glance, we need to determine an appropriate order of visits to these twenty locations; later parts will discuss more realistic conditions, such as working time and access by foot.

The current task can be classified as a search task since the program should pick one of numerous possible routes. Given the fact that the starting point is already defined, there are $19!$ permutations, or about 1.22×10^{17} . Evaluating all possible options would take too much time. Local search techniques are suitable for this task because we are interested in the final option, not the path we take to reach it (Russell & Norvig, 2022). Besides, this technique allows us to compare two Module 2 techniques conveniently.

Purpose and project questions

The project examines which of the two AI heuristics performs better in a fair experimental setting.

- - Do SA and GA create a valid closed tour visiting all 20 places exactly once?
- - Which heuristic leads to shorter routes on average, and not only in the best possible case?
- - What role do reliability and running time play in the final recommendation?
- - Can the chosen algorithm deal with an unknown place without being altered?

Solution overview

The notebook contains the coordinate table, calculates Haversine distance matrix, creates nearest neighbour baseline, performs simulated annealing (SA) and genetic algorithm (GA), performs comparison between thirty runs for both methods and, finally, generates one additional location. The output is stored in the notebook in order to make possible checking the correctness of the data, parameter setting, routing and results without performing the code again.

Correctness is vital; any route, which skips some location or visits it twice, is disregarded. From all remaining valid routes, those with lower sum of distance, variability and computational time are favoured. Evaluation of all those factors together avoids selection of the method according to one successful run only.

Purpose of research: recommendation of the method which gives the best trade-off between the quality, consistency, speed and transparency of routing for 20-location prototype.

2. Problem Formulation and Responsible Scope

Optimisation formulation

Component	Application definition
Search state	A permutation of location indices beginning with depot 0.
Initial state	A nearest-neighbour route perturbed for SA; a mixed greedy/random population for GA.
Successor	A route created by reversing a segment; GA also recombines two parent orders.
Feasible region	All depot-fixed permutations containing every index exactly once.
Objective	Minimise the sum of successive pairwise distances plus the final return edge.
Goal test	No fixed goal state; retain the valid route with the lowest observed objective.

For a given sequence $r = (r_0, r_1, \dots, r_{19})$, where r_0 is fixed as the starting point, Town Hall, the goal is the sum of $d(r_i, r_{i+1}) + d(r_{19}, r_0)$ for all adjacent locations. The distance between two points, d , is based on latitude and longitude coordinates, in radians. Such a formulation of the problem ignores directionality and implies that each pair of stops is mutually accessible. It is done to ensure that we have a discrete, deterministic, static, fully observable, and single-agent state space.

This problem is set up in a similar way as an optimization task rather than a navigational problem. There is no need to return the sequence of intermediate states of the search process; only the optimal path matters. Both simulated annealing and genetic algorithm are able to perform the search in the space of permutations of a given problem with limited amount of memory, while being aware that the result will be suboptimal (Russell & Norvig, 2022).

Scope and responsible-AI boundaries

The data set consists of names only of public places along with their coordinates; therefore, the level of privacy is very low. The problem is that this calculated distance might be considered more accurate than it really is, and it could be misinterpreted. Haversine formula for distance calculation does not consider any other elements like roads, stairs, restricted areas, time period for which place is open, and the amount of time spent there on the way to the destination.

Risk	Control in this prototype
False optimality	Use 'best observed', report 30-run variation, and avoid claiming a proven global optimum.
Unsafe/inaccessible route	Require human approval and replace straight-line distance with verified network travel time before deployment.
Data drift	Freeze inputs for assessment, record source/retrieval date, and revalidate sites operationally.
Non-reproducible result	Embed data, publish parameters, retain outputs and record every random seed.

Responsible-use position. *A short calculated tour is evidence for a planning recommendation, not permission to ignore local access rules or worker judgement.*

3. AI Techniques

3.1 Simulated annealing

Simulated annealing is a stochastic local search heuristic acting on one solution, which is based on controlled cooling techniques used in real annealing processes. At each step, the algorithm chooses two nodes not equal to the depot and reverses the tour between them. If the length of the tour is decreased, the new tour is accepted. Otherwise, if the length is increased by delta kilometers, the new tour is accepted with the probability of $\exp(-\text{delta}/T)$. This random acceptance of decreasing moves helps avoid local minimums. With the geometric decrease of the temperature T , the probability of making a bad move decreases and this leads to the transition from exploration to exploitation (Kirkpatrick et al., 1983).

The reason for using SA in our case is that one tour can be encoded in one array and every neighbor is a valid permutation. Thus, the memory requirement of the heuristic is very low. One of the disadvantages is that the method is sensitive to the temperature parameter and the cooling schedule since fast cooling might result in staying at the mediocre solution while slow cooling will increase the time of execution. For this reason, the heuristic keeps track of the best route found and uses the empirical temperature parameters.

3.2 Genetic algorithm

A genetic algorithm is an evolutionary search process that uses populations. Chromosomes represent the valid permutations of visit orders. Tournament selection favors reproduction of the individuals having shorter tour lengths. Order crossover is used where one parent is used to copy a sequence while using the second parent to fill up the rest in the order they appear without duplication. Inversion mutation involves flipping the sequences at a certain probability to ensure diversity. Elitism involves passing on the best chromosomes without modifications (Holland, 1992).

This genetic algorithm applies since it has diversity in its populations to explore various areas of the route space. The disadvantages include more time taken to compute results, having many parameters to tune, and having the tendency to converge prematurely when selection pressure and elitism reduce diversity too quickly. Validity-preserving operators are very important since numeric crossover will not work.

3.3 Expected trade-off

Criterion	Simulated annealing	Genetic algorithm
Search memory	One current route plus best route	Population of routes
Exploration	Occasional worse moves controlled by temperature	Crossover and mutation across diverse chromosomes
Key controls	Initial temperature, cooling rate, iterations	Population, generations, mutation, elitism, tournament size
Expected strength	Fast, simple replanning	Stable high-quality solutions
Expected failure	Freezes in a local minimum	Population converges prematurely

Fair comparison. Both techniques use the same locations, distance matrix, validity rule, depot, seed set and reporting metrics; only the search mechanism differs.

4. Implementation: Data, Preprocessing and Environment

4.1 Data source and selection

Twenty public places in the downtown area of Sydney were chosen to give an example of a short inspection route. The latitude and longitude values were manually gathered from Google Maps on 3 September 2026 (Google, 2026). The location "Town Hall" is defined as index 0, which is a fixed depot. Every row contains place name, latitude, and longitude values. The table was created specifically for this assignment and does not come from Kaggle, nor does it have any relation to any other solution notebook or code available. This format provides me the ability to clearly show my problem definition and solution approach. In practice, I could use a data management tool, such as the City of Sydney open data portal (City of Sydney, 2026).

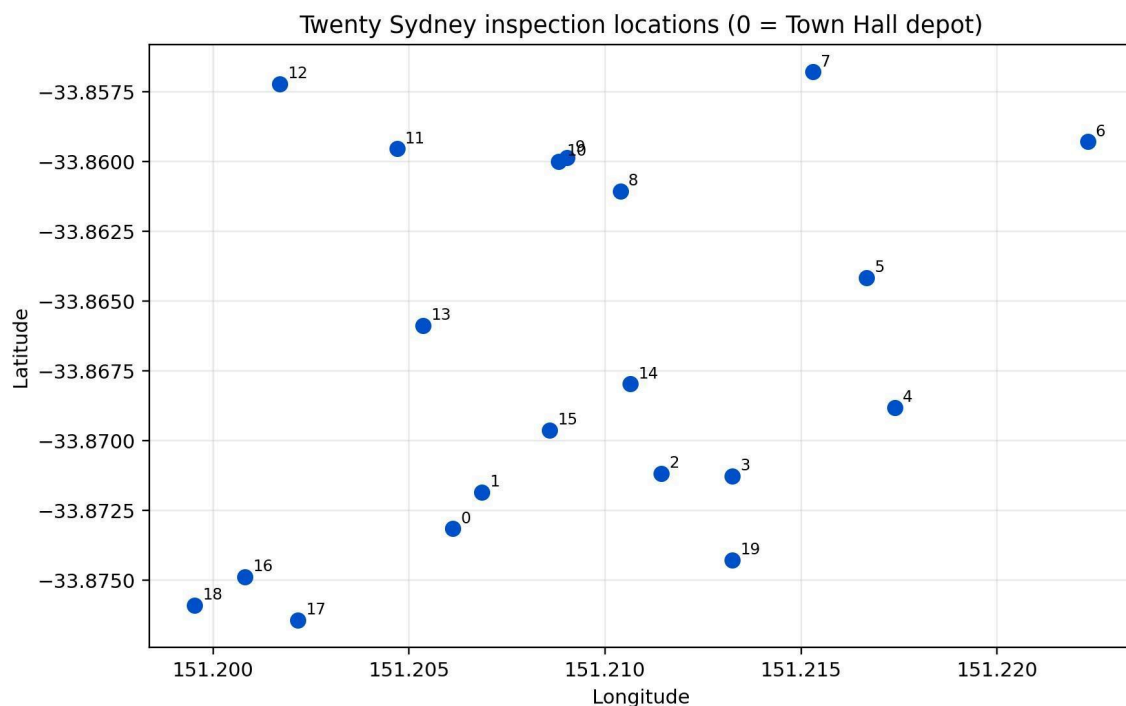


Figure 1. Twenty indexed inspection locations; 0 is the Town Hall depot.

4.2 Preprocessing and validation

A total of 20 unique names, 20 numeric coordinates, and zero missing values have been verified. Prior to distance calculation, the latitude and longitude were converted to radians. A 20x20 Haversine distance matrix was developed, verifying its symmetry and a diagonal of zeroes. The index number 0 was reserved for the depot, and permutation-preserving operators were utilized to make sure each node is represented only once. Coordinates and matrices are stored separately with no need for scaling and imputation.

4.3 Programming environment

The database included twenty unique names and associated twenty numeric coordinates that have been checked in detail, and no missing data has been found. Before computing distances, latitude and longitude have been transformed into radians. Distance matrix of 20x20 elements based on Haversine distance was built, the symmetry of which and zeros on the diagonal were checked. Index 0 is used for the depot location, and permutation-invariant operators were used to guarantee unique representation of each node.

5. Experimental Design and Parameter Justification

5.1 Evaluation protocol

The small parameter experiment from the last comparison was separately evaluated. There were four different settings for each of the two methods being compared tested with five pilot seeds. The setting that resulted in the lowest average distance was used to further increase the search budget for each of the two methods along with the 30 final seeds, (202600-202629). This helps decrease the dependency on the initial lucky route in the comparison.

Method	Config	Mean km	SD km	Best km	Time s
SA	SI-1	8.936	0.081	8.896	0.053
SA	SI-2	8.896	0.000	8.896	0.053
SA	SI-3	8.990	0.126	8.896	0.053
SA	SI-4	8.942	0.078	8.896	0.055
GA	GE-1	9.124	0.474	8.912	0.808
GA	GE-2	8.912	0.000	8.912	0.822
GA	GE-3	8.912	0.000	8.912	1.139
GA	GE-4	8.912	0.000	8.912	1.168

Table 1. Five-run pilot results. Lowest mean within each technique determined the selected setting.

5.2 Locked confirmatory settings

Technique	Selected parameters	Justification
SA	T0 = 0.75; cooling = 0.997; 5,000 iterations; Tmin = 10^{-5}	Pilot SA-2 had the lowest mean and zero pilot spread; 5,000 iterations allow the schedule to cool fully.
GA	Population = 80; 260 generations; mutation = 0.30; elite = 4; tournament = 4	Pilot GA-2 matched the best GA mean at lower runtime than population 120; the larger generation budget confirms convergence.

5.3 Measures

- Primary: best, mean, median and standard deviation of closed-tour distance in kilometres.

Reliability: proportion of the runs that are within 1% of the best distance recorded with either technique.

- Efficiency: average elapsed time for an execution on the same execution environment.

Practical value: Percent gain in mean distance compared to nearest neighbour.

Convergence: best distance achieved to date, in terms of iterations or generations; axes are algorithm specific and not considered equal work units.

The pooled-best route is just the best of the above runs; neither method is a mathematical proof that it is the best route. The length of a result, any unique locations, the depot position and full index coverage are verified prior to the result being used.

Decision rule. Propose a method that has an objectively improved mean distance, reliability and is still a reasonable runtime for planning. Include the quicker alternative in the report if the objective of the operation changes.

6. Findings and Results

6.1 Baseline and confirmatory outcomes

Nearest neighbour produced a valid 12.344 km route. Both AI techniques found the pooled-best 8.896 km tour, a strong indication that each implementation can escape the greedy baseline. Their distributions, however, differ substantially.

Technique	Best km	Mean km	Median	SD	Within 1%	Time s
Simulated annealing	8.896	8.972	8.912	0.206	80%	0.085
Genetic algorithm	8.896	8.911	8.912	0.003	100%	1.121

Table 2. Confirmatory results from 30 independent runs per technique.

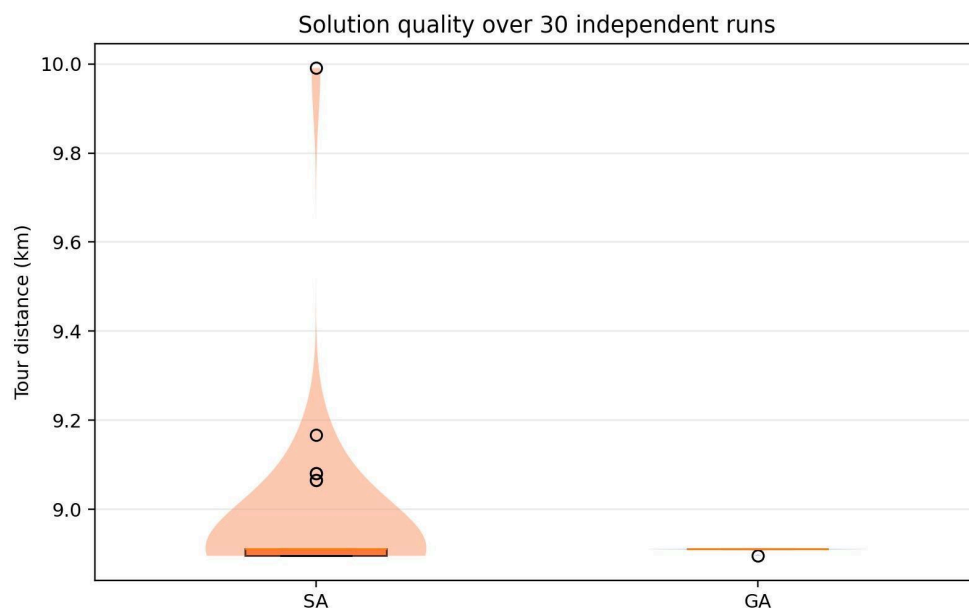


Figure 2. GA results are tightly concentrated; SA includes several longer-route outliers.

6.2 Interpretation of numerical findings

The GA mean (8.911 km) is 0.061 km shorter than the SA mean (8.972 km). There is a greater difference in variation, which is the standard deviation, 0.003 km for GA compared to 0.206 km for SA. The six runs performed in the SA group are outside of the 1% of the pooled best, while all GA runs are inside. This means that the GA is more successfully able to give a strong route.

SA is much faster than GA with an average of 0.085 seconds versus the GA set at 1.121 seconds. In this saved run the GA is approximately 13.2 times slower for this reason, but both are acceptable for making 1 plan per day. The average improvement is 27.3% for SA and 27.8% for GA when compared to nearest neighbour.

Key finding. Equal best distances are not necessarily equal techniques. The GA is much more stable over runs while the main good thing about SA is that it is fast.

7. Comparative Analysis

Best closed tours found across 30 independent runs

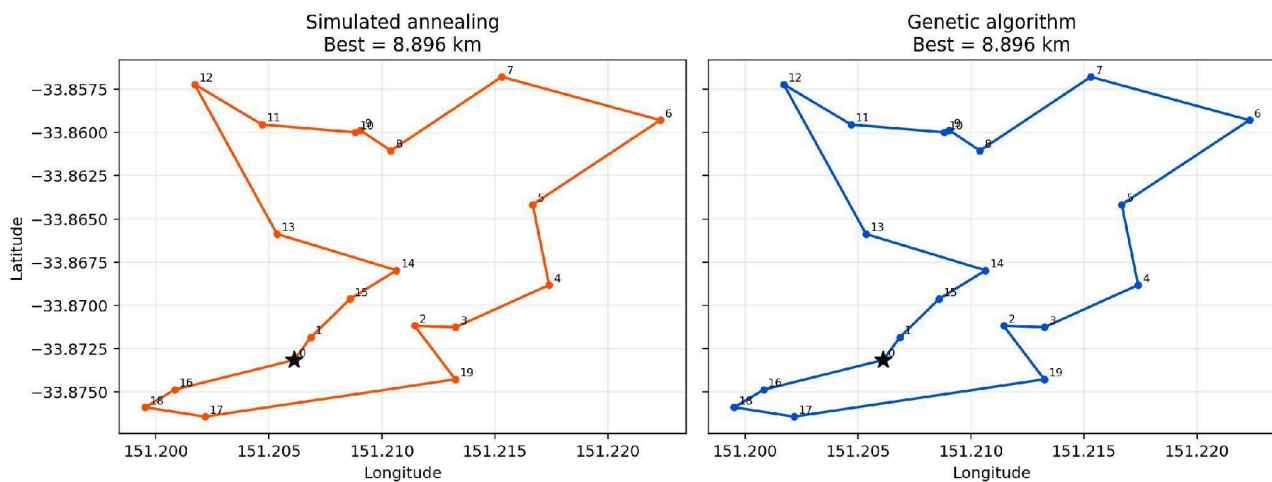


Figure 3. Both methods found the same closed tour in opposite traversal directions; these are equivalent in a symmetric distance matrix.

7.1 Solution quality and robustness

The distance in this symmetric matrix is the same for the best SA and GA tours since they use the same edges, but in opposite directions. This is a good test of the implementations. The difference is evident in each of the 30 runs. The good route segments from the GA population can be preserved and selected by combining them; the bad segments are discarded. The SA population may keep and recombine good segments of routes and sometimes get stuck on a longer route after cooling. Quality is better than a best shot for one planned circuit per day.

7.2 Efficiency and complexity

SA keeps a current route and a best route and has one neighbour checked at every step, so it completes quickly. The GA needs more time and memory to evaluate 80 routes over 260 generations and to do Selection and Crossover. The actual times are dependent on computer and settings. In this case, the secondary is the runtime; both methods are fast enough for normal route planning.

7.3 Contextual strengths and weaknesses

Criterion	SA assessment	GA assessment	Decision
Best observed quality	8.896 km	8.896 km	Tie
Average quality	8.972 km	8.911 km	GA
Consistency	SD 0.206; 80% robust	SD 0.003; 100% robust	GA
Runtime	0.085 s	1.121 s	SA
Implementation	Compact, three main controls	More operators and validity safeguards	SA
Scalability strategy	Increase iterations/restarts	Parallel populations/evaluations possible	Context-dependent

On the stated objective, the GA wins the criteria that directly affect route quality and reliability. SA wins speed and simplicity. If routes had to be recomputed continuously after disruptions, the weighting could reverse; the current recommendation is conditional on a daily or occasional planning context.

Comparative conclusion. Choose GA for dependable planning quality. Choose SA when a near-immediate answer is worth a small and less predictable distance penalty.

8. Recommendation, New Data and Responsible Deployment

8.1 Recommended technique

For this particular example, with 20 locations, I think the genetic algorithm is the best choice. The mean distance of its route is 8.911 km with a standard deviation of just 0.003 km and all 30 runs are within 1% of the best run. It adds approximately 1 second of runtime when an one-time-a-day route is planned, which is not significant. The recommendation is based on all runs, not the common best route.

The data, settings and seed list should be saved together. A subsequent version may be able to execute multiple GA seeds, and present a shortest valid path to a planner. However, in the event that an immediate route recalculation is required, SA should still be retained as a quick alternative.

8.2 Result on future/new data

A 21st location was added, UTS Tower. Seeds 9900-9909 and their original settings were used and both methods were rerun ten times. Not changed any preprocessing, search code. This does a full comparison on new data, not just a reproduced route.

Method	Best km	Mean km	SD km	Mean time (s)
Simulated annealing	10.177	10.245	0.087	0.205
Genetic algorithm	10.280	10.295	0.049	2.615

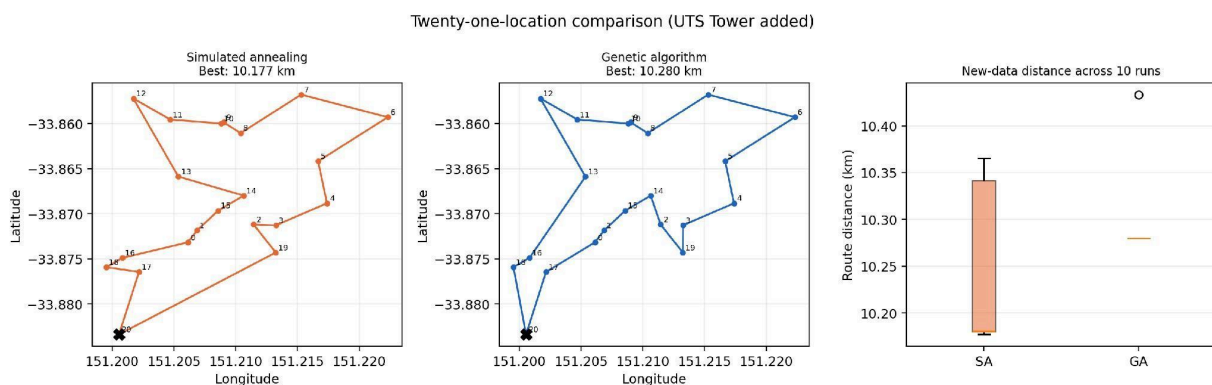


Figure 4. New-data comparison. The black X marks UTS Tower; the box plot shows ten runs per method.

SA registered the smallest mean distance (10.245 km) and was the fastest while GA had the lowest spread (0.049 km). Given this new case, the blanket GA recommendation is weakened; however, the new results show that SA is the more effective approach on distance and speed for this set of 21-locations input.

8.3 Deployment controls

- Replace great-circle distances with accessible pedestrian-network travel times and add visit/service duration.
- Validate opening hours, access and safety rules, and reject any invalid route.
- Show assumptions and keep human approval and an easy manual override.

9. Limitations, Conclusions and Future Work

9.1 Limitations

The primary drawback is Haversine distance is not an accurate measure of walking or driving time. A short straight-line order can be impractical on streets, on the edges of harbours, on hills, crossing them and in restricted areas. The experiment also makes the following assumptions: There is exactly one worker, there is exactly one depot, the service time is the same at each stop, there are no delays, and there are no time windows. A more complex model of vehicle routing or scheduling of a real system would be needed.

Due to typed and handwritten coordinates a centre of a coordinate or a place might not be exactly correct. This is not an official asset register, the rows were checked and all places are public. Also, it will take a different amount of time based on computer and exact code. There are just 30 runs needed for this small comparison, but it doesn't demonstrate what the methods will do on all of the bigger route problems.

The best pooled is NOT an optimal proof. If both of the routes are the same then there is some confidence, but a shorter route could be found. Thus, the term 'best observed' is used, instead of 'optimal route' in this report.

9.2 Conclusions

This project involves transforming a 20 location circuit across Sydney into a search problem and giving two working solutions. Both SA and GA performs better than the nearest neighbour baseline of 12.344 km and both generate an 8.896 km best-observed route. In repeated runs, the mean of the GA is 8.911 km, with very little variation, and each run is within 1% of the best. SA is quicker, but not as consistent. The GA is thus the default, while SA is retained should the need for speed outweigh the need for accuracy.

Add UTS Tower and retest to see the same code can accept another row and compare again. The mean route achieved by SA (10.245 km) was shorter than that of GA (10.295 km) but the latter was not as variable. This indicates that technique might vary depending on the data. The application is still a prototype since distance measure does not consider the actual road network and working conditions.

9.3 Future work

Use an open pedestrian-routing graph or travel-time API and compare with observed travel times.

- Include opening times, service time, load capacity and multiple workers.

Test the scalability using nested evaluation with multiple route instances and a number of locations.

Compare restart policies and hybrid improvement, e.g. apply 2-opt to the best GA chromosome.

- Develop an interface that shows the ordered stops, assumptions, route-validity checks and a manual override. •

Final conclusion. The GA is the more robust solution to the planning goal as it maps out consistently high quality routes from the same set of comparable best case performers.

10. Individual Reflection

Thinking process and decisions

First I wanted to determine what a decent comparison would look like before coding the two algorithms. It would not be fair to have just one handsome track. Both methods required the same input, objective and checks and required more than one run. The type of inspection route was chosen to be the TSP style since the route is clearly defined by the state, neighbour and objective and simulated annealing and genetic algorithms were part of the subject. When Town Hall remains as the depot, the routes are more easily comparable.

I chose to include consistency as a part of solution quality. Both methods involve randomness, so I took a small 5-parameter trial to isolate it from the 30 runs that are left in the final run. The nearest-neighbour baseline helps to make the improvement clearer. Also, I found the term 'pooled best observed' suitable, as there is no way either method can show that its route is the true optimum.

Actions, Barriers and resolution strategies

The first issue was to make a crossover without creating an invalid route. The mixing positions can repeat a position and remove another position, so I applied order crossover and verified the route length, unique indices and fixed depot. Another was the fair selection of settings. I selected a small parameter table, picked the lowest mean for each method and fixed those parameters for the final comparison.

Geographical distance is also not correlated with actual travel. I used Haversine distance as it was easy to understand and implement in a mini application but mentioned the limitation in the report. To address the new-data requirement, I wrote a version of the distance function and a search function that were not dependent on exactly 20 rows, then I ran both functions without a retune, adding UTS Tower in the process.

Lessons learned and achievements

Two working optimisation methods are included in the final notebook: repeatable experiments and saved outputs. The most useful lesson was that two methods may have the same best route and have very different reliability. The GA and SA were tied for best distance but the distribution and within-1% result still provided a clear recommendation. Also, runtime is important: The GA was much slower in absolute terms, but fast enough for a daily plan.

This task helped me understand the importance of determining the evidence before the end of the experiment. The route map never became the only result, thanks to writing the validity checks and comparison measures early. It also had me think more carefully about the distinction between an optimal route that has been found and a route that is mathematically proven to be optimal.

The things I would change:

Next time I would use pedestrian-network travel times, experiment with some different route layouts and compare predictions with actual travel data. I would also add small unit tests for each search operator and test settings around the selected values. There might be a simple interface that displays the ordered stops and assumptions and can still be used to edit the route by a planner. The changes would make the method more useful without making it difficult to understand.

Reflection takeaway. The biggest refinement of my process was going from 'find a short route' to 'make a fair, repeatable decision about which technique to trust'.

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Source and reproducibility note

All third-party concepts, data services and software named in the report are acknowledged above. The coordinate snapshot used for the assessment is reproduced in Appendix A and embedded directly in the notebook. The notebook

records parameters, random seeds and outputs. Google Maps coordinates should be revalidated before operational reuse because map data and place centroids can change.

Appendix A: Input Data

Table A1 reproduces every coordinate embedded in the notebook. Index 0 is the fixed depot.

Index	Location	Latitude	Longitude
0	Sydney Town Hall (depot)	-33.873157	151.206115
1	Queen Victoria Building	-33.871843	151.206851
2	Hyde Park North	-33.871176	151.211440
3	St Mary's Cathedral	-33.871263	151.213253
4	Art Gallery of NSW	-33.868821	151.217392
5	Royal Botanic Garden	-33.864167	151.216667
6	Mrs Macquarie's Chair	-33.859289	151.222311
7	Sydney Opera House	-33.856784	151.215297
8	Circular Quay	-33.861052	151.210390
9	Museum of Contemporary Art	-33.859860	151.209029
10	The Rocks Discovery Museum	-33.859992	151.208817
11	Sydney Observatory	-33.859540	151.204692
12	Barangaroo Reserve	-33.857213	151.201708
13	Wynyard Park	-33.865873	151.205352
14	Martin Place	-33.867974	151.210642
15	Pitt Street Mall	-33.869630	151.208585
16	Darling Harbour	-33.874880	151.200820
17	Chinese Garden of Friendship	-33.876436	151.202163
18	ICC Sydney	-33.875899	151.199522
19	Australian Museum	-33.874279	151.213240

Data checks

- Rows: 20; unique location names: 20; missing values: 0.
- Distance matrix shape: 20 x 20; symmetric: true; diagonal zero: true.
- Earth radius used in Haversine calculation: 6,371.0088 km.
- Future test row: UTS Tower, -33.883360 latitude, 151.200541 longitude.

Validity invariant

Every candidate route must contain exactly the integer set $\{0, 1, \dots, 19\}$, with 0 in the first position. The closing return to 0 is applied only by the objective function and is not duplicated inside the chromosome. The future test changes the required index set to $\{0, 1, \dots, 20\}$ without changing the invariant logic.

Appendix B: Best Route and Rubric Traceability

B1. Best observed GA route

Step	Location
0	Sydney Town Hall (depot)
1	Darling Harbour

2	ICC Sydney
3	Chinese Garden of Friendship
4	Australian Museum
5	Hyde Park North
6	St Mary's Cathedral
7	Art Gallery of NSW
8	Royal Botanic Garden
9	Mrs Macquarie's Chair
10	Sydney Opera House
11	Circular Quay
12	Museum of Contemporary Art
13	The Rocks Discovery Museum
14	Sydney Observatory
15	Barangaroo Reserve
16	Wynyard Park
17	Martin Place
18	Pitt Street Mall
19	Queen Victoria Building
20	Sydney Town Hall (depot)

Closed-tour distance: 8.896 km. The reverse traversal is equivalent under the symmetric matrix.

B2. Rubric evidence map

Rubric area	Direct evidence
Referencing	APA in-text citations and complete reference list on page 13
Executive summary	Purpose, techniques, major findings, conclusion and recommendation on page 3
Introduction	Problem, context, need, questions and solution overview on page 4
AI techniques	Mechanisms, application relevance and limitations on pages 5-6
Implementation	Source, preprocessing, environment, parameters, experiments and outputs on pages 6-9
Comparison	Quality, reliability, runtime, strengths, weaknesses and conditional recommendation on page 10
Conclusions	Findings linked to objective, limitations and future work on page 12
Working solution	Self-contained .ipynb plus executed-output notebook PDF
Annotation	Original explanation precedes every code cell
Reflection	Decisions, actions, barriers, resolutions, learning and improvements on pages 12-13